

Genetic Diversity and Extinction Risk in Animal Species

A Mann–Whitney Analysis of COI Nucleotide Diversity

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Introduction

Low genetic diversity is expected to make populations less able to adapt to disease and environmental change, and therefore more vulnerable to extinction. Large public datasets now allow us to test this idea across many species. In this report we address a simple question: **Is genetic diversity associated with extinction risk in animal species?** The hypothesis is that species classified as threatened by the IUCN Red List have lower genetic diversity than non-threatened species.

Data Sample

The dataset [1] is the supplementary table of Petit-Marty et al., which reports the nucleotide diversity (π) of the mitochondrial COI gene for 4,363 animal species, together with their IUCN Red List status. The table was distributed as a PDF and converted to a CSV file. The relevant columns are:

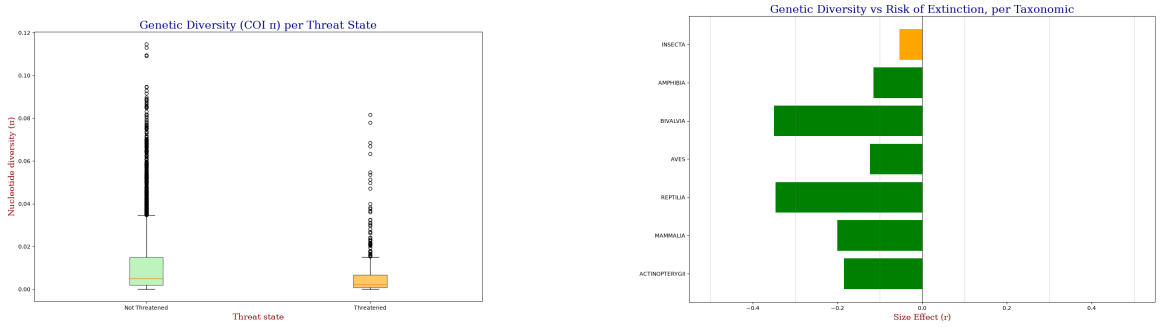
- `class`: taxonomic class (e.g. Mammalia, Aves, Insecta),
- `red_list_status`: IUCN category (LC, NT, VU, EN, CR, ...),
- `ntd_div_pi`: nucleotide diversity π of the COI gene.

Species were grouped as *Threatened* (VU, EN, CR) or *Not Threatened* (LC, NT). Six species with non-classifiable codes were removed, leaving 4,357 species (383 threatened, 3,974 not threatened). No missing values or duplicates were found.

Statistical Analysis. The values of π are strongly right-skewed, so we used the non-parametric Mann–Whitney U test, implemented manually in Python (`pandas`, `matplotlib`) with a normal approximation for the p-value. The effect size was computed as $r = z/\sqrt{N}$ and interpreted with Cohen's thresholds (0.1 small, 0.3 medium, 0.5 large) [2]. The test was repeated within each taxonomic class having at least 20 species in both groups.

Results and Discussion

Figure 1 compares both groups. The median diversity is 0.0051 for non-threatened and 0.0023 for threatened species, less than half. The difference is highly significant ($U = 560{,}904$, $z = -8.51$, $p < 0.001$), but the effect size is small ($r = -0.129$).



(a) COI nucleotide diversity by threat status.

(b) Effect size r per class.

Figure 1: Distribution of genetic diversity and subgroup effect sizes (r). Green: $p < 0.05$; orange: not significant.

Figure 1b and Table 1 show the same test per taxonomic class. The effect is negative in all seven classes and significant in six. It is strongest in Bivalvia and Reptilia ($r \approx -0.35$) and negligible in Insecta ($r = -0.053$, $p = 0.26$).

Table 1: Results per class (n_T, n_N: threatened and not threatened species).

Class	n_T	n_N	p**-value**	r
Actinopterygii	112	1,313	2.9×10^{-12}	-0.185
Mammalia	57	702	3.4×10^{-8}	-0.200
Reptilia	45	190	1.1×10^{-7}	-0.347
Aves	52	1,000	6.4×10^{-5}	-0.123
Bivalvia	29	73	4.1×10^{-4}	-0.350
Amphibia	54	287	0.034	-0.115
Insecta	34	409	0.260	-0.053

Threatened species therefore show significantly lower genetic diversity, which fits the idea that small or declining populations lose variation through drift, bottlenecks and inbreeding. The small effect size indicates that threat status explains only part of the variation in π, as expected since extinction risk also depends on habitat loss, hunting or pollution, and a single mitochondrial gene captures only part of a species' diversity. The result is an association, not proof of causation. Seven class-level tests were run without correction; with a Bonferroni threshold of $0.05/7 \approx 0.007$, Amphibia would no longer be significant.

References

[1] Petit-Marty, N., Vázquez-Luis, M., & Hendriks, I. E. (2020). *Use of the nucleotide diversity in COI mitochondrial gene as an early diagnostic of conservation status of animal species.* Conservation Letters, 14(1). doi:10.1111/conl.12756.

[2] Sullivan, G. M., & Feinn, R. (2012). *Using effect size—or why the P value is not enough.* Journal of Graduate Medical Education, 4(3), 279–282. doi:10.4300/jgme-d-12-00156.1.